BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 107 · Trace Tables

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a diamond represent in a flowchart?

Q2. In OCR ERL, which value can random(1,6) return?
A. 0
B. 7
C. 6
D. 1.5

Q3. “A USB flash drive stores data on a rotating magnetic platter.” Correct this. Explain why.

Q4. A large file must cross a slow link. “Compression always makes files larger.” Correct this. How might compression affect transfer time?

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I Do · Read new knowledge

Key vocabulary

Trace table — Records program variables and values as instructions execute.

Accumulator — Variable maintaining a running total.

Iteration counter — Variable tracking loop repetitions.

Assignment — Instruction replacing a variable's stored value.

New learning

Read statements in order: A trace records program state after each instruction that changes a variable.

Track an accumulator: An accumulator holds a running total; it changes each time a value is added.

Include selection and loops: Trace iteration counters, condition outcomes and updated values; do not record an update when a branch is skipped.

Check final output: The final displayed result should agree with the last recorded relevant value.

Worked example

Trace total = 0; for i = 1 to 4; total = total + i; next i; print(total). [5 marks]

1. Initialise variables

2. Before first iteration, total = 0.

3. After i=1,2,3,4 the totals are 1, 3, 6, 10.

4. The output is 10; rows record each update to total. [5]

We Do · Practise with guidance

Guided 1

Trace total=0; for i=1 to 3; total=total+2; next i. Give each value. [4]

Worked method: After i=1 total 2 [1], i=2 total 4 [1], i=3 total 6 [1], final output 6 [1].

Guided 2

Trace x=8; if x>5 then x=x-3 endif; print(x). [3]

Worked method: Condition 8>5 is true [1]; x becomes 5 [1]; output 5 [1].

Guided 3

Explain why trace tables can expose logic errors in a loop. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: They show values after each iteration [1], letting a programmer see where an unexpected value appears [1], compare with intended behaviour [1], and locate the faulty update or condition [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the loop statement that adds every file size.

```
array files = [4,6,10]
total = 0
for i = 0 to 2
    ...
next i
print(total)
```

write

Question 1

Trace x=2; x=x*3; x=x-1; print(x). Give x after each update and the output. [3] Source: Original OCR J277-style practice question.

Question 2

Trace total=0; for i=1 to 3; total=total+i; next i. Record three totals. [3] Source: Original OCR J277-style practice question.

Question 3

Explain why a trace must record a variable only after the relevant assignment has executed. [2] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A trace table can help check how values change during a program.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A decision or condition with alternative paths.

Q2. C — 6

Q3. Solid-state storage with no moving read/write mechanism.

Q4. A smaller file uses less storage and can be sent faster over the same link. Compression and decompression take processing time.

I Do / We Do pages · Guided practice

Q1. After i=1 total 2 [1], i=2 total 4 [1], i=3 total 6 [1], final output 6 [1].

Q2. Condition 8>5 is true [1]; x becomes 5 [1]; output 5 [1].

Q3. They show values after each iteration [1], letting a programmer see where an unexpected value appears [1], compare with intended behaviour [1], and locate the faulty update or condition [1].

You Do pages · Code First

```
array files = [4,6,10]
total = 0
for i = 0 to 2
    total = total + files[i]
next i
print(total)
```

Expected result: 20

You Do pages · Independent questions

Q1. After multiplication x=6 [1]; after subtraction x=5 [1]; output 5 [1].

Q2. 1 [1], 3 [1], 6 [1].

Q3. Before the assignment the old value is held [1]; recording afterwards shows the new program state correctly [1].

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: